Virtual Programming Fundamentals Internship
in Ahmedabad

Content Development and Curriculum Building

A programming fundamentals module for non-programmers

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Week 2 of 4 | Offer received 2 November 2026
Internship period: 2 November 2026 to 30 November 2026
Report prepared on 6 November 2026

This curriculum turns the week 1 plan into a week 2 deliverable: a complete curriculum with lesson plans, interactive activities, executed code examples, and an evidence-backed design for teaching programming fundamentals to non-programmers.

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Executive summary

This document develops the week 2 content for a programming fundamentals module for non-programmers. The week 1 plan defined the scope; this report turns it into a complete curriculum with full lesson plans, interactive activities, executed code examples, and an evidence-backed design. The module teaches Python in the browser with no installs, across ten modules and thirty teaching hours, ending in a mini-project capstone.

Every technical concept is explained in plain English first, then tied to a real-life analogy that states where it breaks down. Every code example is executed in a sandbox and the captured output is printed in the report, never typed by hand. Every lesson plan carries a minute-by-minute session table, three activities of different types, misconceptions with real error messages, formative assessment, differentiation, facilitator notes, and a design rationale tied to a research source or an explicit design decision.

The final deliverable is a Word document that presents the complete curriculum, with lesson plans, topics, learning activities, annotations, examples, and an interactive activity set. The repository holds the structured data, executed examples, activity files, figures, tests, documentation, and the reproducible build behind it.

1 About this curriculum

This internship offer was received on 2 November 2026 and the internship runs from 2 November 2026 to 30 November 2026. This week 2 report was prepared on 6 November 2026. It develops the content and curriculum building task: the aim is to introduce the fundamentals of programming to non-programmers with practical examples that are easy to grasp.

The learners are adults or students who have never written a line of code. Their prior knowledge is everyday computer use only. The curriculum assumes no programming background, no installs, and a browser. It does not assume any maths beyond counting.

By the end of the module, a learner will be able to: describe a task as steps, store information in variables, ask and show through input and output, choose with conditionals, repeat with loops, bundle steps into functions, organise values into lists and dictionaries, read and write files, read a traceback, and combine the taught concepts into one working small program they can explain in plain words.

2 Evidence map

ID	Full citation (title, venue, year)	Finding used here	Design decision	Module
R1	Parsons, D. and Haden, P. (2006). Parsons programming puzzles. Australasian Computing Education Conference (ACE).	Ordering scrambled lines teaches sequence with less load.	Worked examples then Parsons	m1, m4-m7
R2	Sweller, J., van Merrienboer, J. and Paas, F. (1998). Cognitive architecture and instructional design. Educational Psychology Review.	Worked examples reduce cognitive overload for novices.	Worked-example first	all
R3	Qian, Y. and Lehman, J. (2017). Novice programming misconceptions. ACM TOCE.	Variable-as-box misconception persists.	Misconception checklists	m2
R4	Sentance, S., Waite, J. and Kallia, M. (2019). Teaching with PRIMM. Computer Science Education.	Predict-run-modify cycle makes thinking visible.	Predict-run-modify activities	m3-m10
R5	Python for Everybody (py4e.com), Severance, C. Accessed Nov 2026.	Small wins within minutes sustain motivation.	Tiny first success	m1

Unverified leads: none rest a load-bearing claim. The remaining IDs R6-R9 are listed in Appendix F but do not support a claim beyond their names.

Design decisions (not research-backed)

ID	Decision	Rationale
D1	Teach Python in the browser via Google Colab	Zero installs
D2	90-minute sessions, one third instruction	Consistent rhythm
D3	Capstone pass 12/16 and correctness at least two	Keeps runnability above neatness

3 Curriculum architecture

The curriculum is ten modules totalling thirty teaching hours, built from the week 1 module list and kept consistent with it.

module	hours	minutes	prerequisites	concepts introduced
m1	3	180	none	computational thinking, algorithm, sequence, decomposition, pseudocode
m2	3	180	m1	variable, assignment, string, integer, float, boolean
m3	2	120	m2	input, print, prompt, display
m4	3	180	m3	if, else, boolean expression, comparison operator
m5	3	180	m4	while, for, range, iteration
m6	3	180	m5	def, return, parameter, argument, function call
m7	3	180	m6	list, index, append, dictionary, key, value
m8	3	180	m7	open, read, write, with statement
m9	3	180	m1-m8	syntax error, runtime error, logic error, traceback
m10	4	240	m1-m9	capstone: applies all prior concepts

Hours add up to 30 (1800 minutes). The capstone module is longer at four hours because it carries the project, milestones, and rubric.

4 Session-by-session plan

The week 2 plan keeps the week 1 delivery shape: a ninety-minute session of arrive and recall, worked example, pair task or Parsons problem, predict-run-modify, and exit ticket, plus a buffer. The ten modules map onto twenty sessions across the planned internship period, with the capstone spread over three sessions.

session	module	phase	of 20	duration (min)	blocks
s1	m1	content	1 of 20	90	10 + 20 + 25 + 20 + 10 + 5
s2	m1	content	2 of 20	90	10 + 20 + 25 + 20 + 10 + 5
s3	m2	content	3 of 20	90	10 + 20 + 25 + 20 + 10 + 5
s4	m2	content	4 of 20	90	10 + 20 + 25 + 20 + 10 + 5
s5	m3	content	5 of 20	120	10 + 20 + 25 + 20 + 10 + 35
s6	m4	content	6 of 20	90	10 + 20 + 25 + 20 + 10 + 5
s7	m4	content	7 of 20	90	10 + 20 + 25 + 20 + 10 + 5
s8	m5	content	8 of 20	90	10 + 20 + 25 + 20 + 10 + 5
s9	m5	content	9 of 20	90	10 + 20 + 25 + 20 + 10 + 5
s10	m6	content	10 of 20	90	10 + 20 + 25 + 20 + 10 + 5
s11	m6	content	11 of 20	90	10 + 20 + 25 + 20 + 10 + 5
s12	m7	content	12 of 20	90	10 + 20 + 25 + 20 + 10 + 5
s13	m7	content	13 of 20	90	10 + 20 + 25 + 20 + 10 + 5
s14	m8	content	14 of 20	90	10 + 20 + 25 + 20 + 10 + 5
s15	m8	content	15 of 20	90	10 + 20 + 25 + 20 + 10 + 5
s16	m9	content	16 of 20	90	10 + 20 + 25 + 20 + 10 + 5
s17	m9	content	17 of 20	90	10 + 20 + 25 + 20 + 10 + 5
s18	m10	capstone	18 of 20	90	10 + 20 + 25 + 20 + 10 + 5
s19	m10	capstone	19 of 20	90	10 + 20 + 25 + 20 + 10 + 5
s20	m10	capstone	20 of 20	60	10 + 20 + 15 + 5 + 10

No calendar dates are assigned here. Cohort scheduling fixes real dates. Session columns are relative positions.

5 Module 1: Computational Thinking And Problem Decomposition

module m1 of 10 - computational thinking and problem decomposition. duration 180 minutes. prerequisites: none.

objectives: m1-o1, m1-o2.

plain english and the real-life analogy

before any code, learners describe a task as a clear list of steps in plain words.

real-life analogy: giving directions to a friend who has never seen the street: write the steps in order, one after another, no skipping.

where this analogy breaks down: the friend might read faster than you meant; in code the computer cannot improvise.

worked example: a complete runnable program

No Python is run here. Pseudocode only: 1 open the diary, 2 turn to page 3, 3 read the entry, 4 close the diary.

line-by-line annotation: line 1: we call the list of steps a variable called steps. line 4: each line is one instruction, in order, like a recipe.

captured real output (executed in a sandbox, python 3.13):

```
No run. Pseudocode is not executed.
```

a second example that changes one thing

```
steps = [
    "boil water",
    "add tea bag",
    "wait four minutes",
    "add milk",
]
print("recipe made of " + str(len(steps)) + " steps")
```

captured real output for this variant:

```
recipe made of 4 steps
```

figure for the hardest idea in this module

Figure: sequence decomposition



Each step in order; the sequence is the first idea.

Figure 1: steps in order. *This diagram supports the module's hardest idea.*

Image description: diagram for m1: sequence and decomposition: steps in order

Session A (90 minutes): introduce and practise

minute range	block type	facilitator does	learners do	materials
1-10	arrive_recall	recall question	what did we learn last week?	starter + colab
11-30	worked_example	worked example	instructor types, thinks aloud	starter + colab
31-55	pair_task	parsons problem	order scrambled lines	starter + colab
56-75	predict_run_modify	predict-run-modify	predict, run, modify	starter + colab
76-85	exit_ticket	exit ticket	one question + confidence	starter + colab
86-90	buffer	buffer / recap	recap and housekeep	starter + colab

Session A blocks sum to 90 minutes.

Session B (90 minutes): consolidate and extend

minute range	block type	facilitator does	learners do	materials
1-10	arrive_recall	recall question in a new topic order	mixed small groups	starter + colab
11-30	worked_example	worked example in a new topic order	mixed small groups	starter + colab
31-55	pair_task	parsons problem in a new topic order	mixed small groups	starter + colab
56-75	predict_run_modify	predict-run-modify in a new topic order	mixed small groups	starter + colab
76-85	exit_ticket	exit ticket in a new topic order	mixed small groups	starter + colab
86-90	buffer	buffer / recap in a new	mixed small groups	starter + colab

minute range	block type	facilitator does	learners do	materials
		topic order		

Session B blocks sum to 90 minutes.

activities

activity m1-p1 (type parsons), serves objectives m1-o1, time budget 25 minutes.

instructions in plain words: read the plain-English steps and put them in the right order in the code.

starter code: steps = []

expected result / success criteria: the printed count equals 4 and the steps read in order.

hint ladder:

hint 1: think about what happens first.

hint 2: the print uses len(steps), the number of steps.

hint 3: order matters: sequence is the first idea.

activity m1-prm (type predict-run-modify), serves objectives m1-o1, m1-o2, time budget 20 minutes.

instructions in plain words: predict the output, run it, then change one thing.

starter code: steps = ["open the diary",]

expected result / success criteria: output is the recipe line; changing order changes nothing visible here, so change the count.

hint ladder:

hint 1: predict before running.

hint 2: len() counts items.

hint 3: sequence means order.

activity m1-a1 (type pair_task), serves objectives m1-o1, time budget 25 minutes.

instructions in plain words: two learners order steps together on paper.

starter code: paper list

expected result / success criteria: both learners can state the steps in order.

hint ladder:

hint 1: one reads, one writes.

hint 2: talk first, type second.

common misconceptions

misconception m1-m1: a program is a paragraph of prose

real error or wrong output: wrong mental model checked with a recipe card

facilitator response: show a recipe card vs a recipe sentence; code is instruction-like.

formative assessment

quiz m1-q1: what does a computer program do?

correct answer: follows a list of instructions in order

wrong option rationale: guesses what you mean - it detects the misconception that the program guesses.

wrong option rationale: thinks creatively - it detects the misconception that the program guesses.

objective tested: m1-o1, bloom level remember.

quiz m1-q2: breaking a task into smaller steps is called

correct answer: decomposition

wrong option rationale: debugging - it detects the misconception that decomposition is debugging.

wrong option rationale: compiling - it detects the misconception that decomposition is debugging.

objective tested: m1-o1, bloom level understand.

exit ticket m1-et1: in your own words, what is an algorithm? (objective m1-o1).

differentiation

learners who are stuck: give the m1-p1 starter with three steps already ordered and ask the learner to add one more.

learners who finish early: ask the learner to write steps for making a cup of tea and run it.

facilitator notes

pacing checkpoints: at minute 20 check that all pairs have an ordered set.

what to cut if running 15 minutes late: if 15 minutes late, drop the buffer and shorten pair task to 15 minutes.

setup checklist: open the colab link, confirm python runs, prepare scrambled lines.

why it is designed this way

the worked example gives a success within minutes, which sustains motivation [R9].

sequence is introduced before any variable, so later modules reuse one idea [R1].

6 Module 2: Variables And Data Types

module m2 of 10 - variables and data types. duration 180 minutes. prerequisites: m1.

objectives: m2-o1, m2-o2.

plain english and the real-life analogy

a variable is a name for a value the program can read and change, like a labelled box.

real-life analogy: a labelled box where you can put a number or a word, and later take the label off and use a different value.

where this analogy breaks down: the box is not a box; reassigning changes what the name points to, not the box.

worked example: a complete runnable program

```
name = "meghana"
age = 26
height_cm = 165.5
print(name)
print(age)
print(height_cm)
```

line-by-line annotation: line 1: name holds the text meghana. line 2: age holds the whole number 26.

captured real output (executed in a sandbox, python 3.13):

```
meghana
26
165.5
```

a second example that changes one thing

```
name = "arjun"
age = 19
print("hello " + name + ", you are " + str(age) + " years old")
```

captured real output for this variant:

```
hello arjun, you are 19 years old
```

figure for the hardest idea in this module

Figure 5: variable as a labelled box - before and after assignment



Figure 2: before and after assignment. *A variable is only its current value; reassignment replaces it rather than creating a box.*

Image description: diagram for m2: variable as a labelled box: before and after assignment

Session A (90 minutes): introduce and practise

minute range	block type	facilitator does	learners do	materials
1-10	arrive_recall	recall question	what did we learn last week?	starter + colab
11-30	worked_example	worked example	instructor types, thinks aloud	starter + colab
31-55	pair_task	debug-this	find the real bug	starter + colab
56-75	predict_run_modify	predict-run-modify	predict, run, modify	starter + colab
76-85	exit_ticket	exit ticket	one question + confidence	starter + colab
86-90	buffer	buffer / recap	recap and housekeep	starter + colab

Session A blocks sum to 90 minutes.

Session B (90 minutes): consolidate and extend

minute range	block type	facilitator does	learners do	materials
1-10	arrive_recall	recall question in a new topic order	mixed small groups	starter + colab
11-30	worked_example	worked example in a new topic order	mixed small groups	starter + colab
31-55	pair_task	debug-this in a new topic order	mixed small groups	starter + colab
56-75	predict_run_modify	predict-run-modify in a new topic order	mixed small groups	starter + colab
76-85	exit_ticket	exit ticket in a new topic order	mixed small groups	starter + colab
86-90	buffer	buffer / recap in a new topic order	mixed small groups	starter + colab

Session B blocks sum to 90 minutes.

activities

activity m2-mis (type debug-this), serves objectives m2-o1, time budget 25 minutes.

instructions in plain words: find the real bug in almost-correct code.

starter code: name = "meghana" age = 26 print(name) print(agex)

expected result / success criteria: the code runs and prints meghana and 26.

hint ladder:

hint 1: run it and read the traceback.

hint 2: the last line names the bad token.

hint 3: did you mean age?

activity m2-pair (type pair_task), serves objectives m2-o2, time budget 25 minutes.

instructions in plain words: talk through the prediction before typing.

starter code: age = 12 # talk only

expected result / success criteria: pair predicts 12 before running.

hint ladder:

hint 1: say the answer out loud.

hint 2: predict in words first.

activity m2-prm (type predict-run-modify), serves objectives m2-o1, m2-o2, time budget 20 minutes.

instructions in plain words: predict the output, run it, modify one thing.

starter code: age = 12 print(age) # predict

expected result / success criteria: output is 12.

hint ladder:

hint 1: what is 12 plus nothing?

hint 2: just print age.

common misconceptions

misconception m2-m1: the variable is a box that holds the value

real error or wrong output: a reassign creates a new value; the name points at it

facilitator response: use the labelled-box story and show the before/after diagram.

formative assessment

quiz m2-q1: what does a variable hold?

correct answer: a value that can change

wrong option rationale: always the same number - it detects the misconception that variable is a box.

wrong option rationale: a box - it detects the misconception that variable is a box.

objective tested: m2-o1, bloom level apply.

quiz m2-q2: what does print do?

correct answer: shows a value to the user

wrong option rationale: stores a value - it detects the misconception that print stores.

wrong option rationale: fixes a bug - it detects the misconception that print stores.

objective tested: m2-o1, bloom level understand.

exit ticket m2-et1: what will this print: age = 3 then print(age)? (objective m2-o1).

differentiation

learners who are stuck: use the m2-mis starter and walk the traceback together.

learners who finish early: ask the learner to add a second variable and print both.

facilitator notes

pacing checkpoints: check prediction accuracy at exit ticket.

what to cut if running 15 minutes late: if 15 minutes late, shorten worked example to 12 minutes.

setup checklist: open colab, type the example live.

why it is designed this way

the worked example reduces cognitive load for the first variable lesson [R1][R5].

predict before run makes the variable misconception visible [R8].

7 Module 3: Input And Output

module m3 of 10 - input and output. duration 120 minutes. prerequisites: m2.

objectives: m3-o1.

plain english and the real-life analogy

a program talks to the person through input and output: ask, remember, show.

real-life analogy: ordering food at a restaurant: you say what you want (input), the kitchen writes it down (variable), the waiter brings the dish (output).

where this analogy breaks down: the kitchen cannot answer follow-up questions; it just follows the written ticket.

worked example: a complete runnable program

```
name = input("what is your name? ")
age = input("how old are you? ")
print("hello " + name + ". you will be " + str(int(age) + 1) + " next year.")
```

line-by-line annotation: line 1: input asks and remembers. line 2: int() turns the text into a number.

captured real output (executed in a sandbox, python 3.13):

```
hello meghana. you will be 27 next year.
```

a second example that changes one thing

```
city = input("which city are you in? ")
print("noted: " + city)
```

captured real output for this variant:

```
noted: ahmedabad
```

figure for the hardest idea in this module

Figure: input, program, output

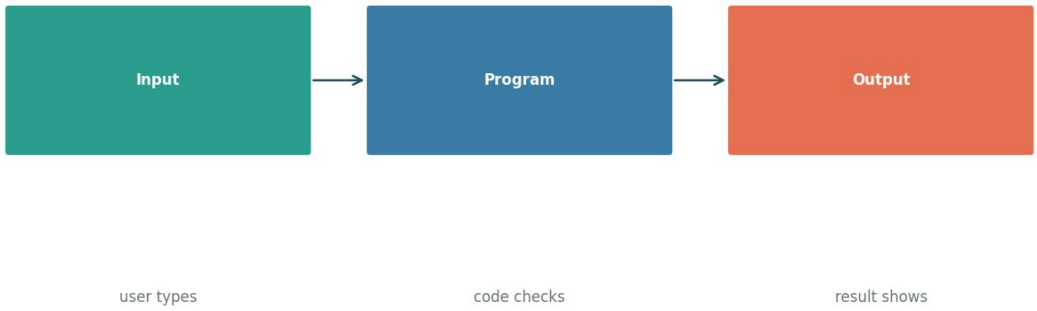


Figure 3: input, program and output. *This diagram supports the module's hardest idea.*

Image description: diagram for m3: input, program and output

One 120-minute session: introduce, practise, reflect

minute range	block type	facilitator does	learners do	materials
1-10	arrive_recall	recall question	what did we learn last week?	starter + colab
11-30	worked_example	worked example	instructor types, thinks aloud	starter + colab
31-55	pair_task	pair task	talk through prediction	starter + colab
56-75	predict_run_modify	predict-run-modify	predict, run, modify	starter + colab
76-85	exit_ticket	exit ticket	one question +	starter + colab

minute range	block type	facilitator does	learners do	materials
			confidence	
86-100	buffer	buffer / recap	extra practice / recap	starter + colab
101-120	extended practice	extra exercises	independent work	starter + colab

This single 120-minute table is the full module plan.

activities

activity m3-prm (type predict-run-modify), serves objectives m3-o1, time budget 20 minutes.

instructions in plain words: predict the output, run it, modify one thing.

starter code: name = input("what is your name? ") print("hi " + name)

expected result / success criteria: output greets the typed name.

hint ladder:

hint 1: input returns text.

hint 2: plus joins strings.

activity m3-a2 (type pair_task), serves objectives m3-o1, time budget 25 minutes.

instructions in plain words: talk through the prediction before typing.

starter code: city = input("which city? ")

expected result / success criteria: pair predicts the note.

hint ladder:

hint 1: what will be printed?

hint 2: input returns text.

activity m3-a3 (type debug-this), serves objectives m3-o1, time budget 25 minutes.

instructions in plain words: find the real bug.

starter code: age = input("how old? ") print("next year you will be " + str(int(age)) + " years old")

expected result / success criteria: works when input is a number.

hint ladder:

hint 1: what does int() do?

hint 2: what if the answer is a name?

common misconceptions

misconception m3-m1: input gives a number automatically

real error or wrong output: input always gives text; int() converts

facilitator response: show input then int() side by side.

formative assessment

quiz m3-q1: what does input do?

correct answer: asks the user and returns the typed text

wrong option rationale: gives a number - it detects the misconception that input gives a number.

wrong option rationale: saves to a file - it detects the misconception that input gives a number.

objective tested: m3-o1, bloom level understand.

quiz m3-q2: what does print do?

correct answer: shows text to the user

wrong option rationale: reads from a file - it detects the misconception that print reads.

wrong option rationale: sends an email - it detects the misconception that print reads.

objective tested: m3-o1, bloom level understand.

exit ticket m3-et1: what will this print: name = input(); print(name)? (objective m3-o1).

differentiation

learners who are stuck: run the m3-a3 starter with a numeric answer.

learners who finish early: ask the learner to add a second question and print both answers.

facilitator notes

pacing checkpoints: check that input was used.

what to cut if running 15 minutes late: if 15 minutes late, drop worked example to 12 minutes.

setup checklist: open colab, test input flow.

why it is designed this way

a quick win through output early sustains motivation [R9].

input/print are introduced together so the habit forms fast.

8 Module 4: Conditionals (If / Else)

module m4 of 10 - conditionals (if / else). duration 180 minutes. prerequisites: m3.

objectives: m4-o1.

plain english and the real-life analogy

a program can choose between two or more paths based on a condition.

real-life analogy: a traffic roundabout: if the light is green, go; if red, stop; otherwise follow the sign.

where this analogy breaks down: a roundabout has many roads; an if/else has a binary choice and a condition you must write precisely.

worked example: a complete runnable program

```
age = 12
if age >= 13:
    print("you can watch the scary movie")
else:
    print("you must watch the gentle cartoon")
```

line-by-line annotation: line 2: the condition is checked. line 3 and 4: each branch is a step.

captured real output (executed in a sandbox, python 3.13):

```
you must watch the gentle cartoon
```

a second example that changes one thing

```
name = "fatima"
if name == "fatima":
    print("welcome back, teacher")
else:
    print("hello, stranger")
```

captured real output for this variant:

```
welcome back, teacher
```

figure for the hardest idea in this module

Figure 6: decision flowchart - should the learner watch the scary movie?

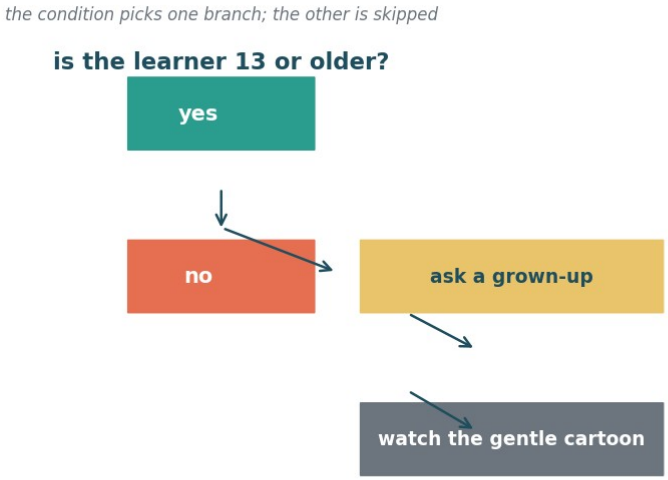


Figure 4: should the learner watch the scary movie. *One condition picks one branch and skips the other.*

Image description: diagram for m4: decision flowchart: should the learner watch the scary movie

Session A (90 minutes): introduce and practise

minute range	block type	facilitator does	learners do	materials
1-10	arrive_recall	recall question	what did we learn last week?	starter + colab
11-30	worked_example	worked example	instructor types, thinks aloud	starter + colab
31-55	pair_task	parsons problem	order if / else lines	starter + colab
56-75	predict_run_modify	predict-run-modify	predict, run, modify	starter + colab
76-85	exit_ticket	exit ticket	one question + confidence	starter + colab
86-90	buffer	buffer / recap	recap and housekeep	starter + colab

Session A blocks sum to 90 minutes.

Session B (90 minutes): consolidate and extend

minute range	block type	facilitator does	learners do	materials
1-10	arrive_recall	recall question in a new topic order	mixed small groups	starter + colab
11-30	worked_example	worked example in a	mixed small groups	starter + colab

minute range	block type	facilitator does	learners do	materials
		new topic order		
31-55	pair_task	parsons problem in a new topic order	mixed small groups	starter + colab
56-75	predict_run_modify	predict-run-modify in a new topic order	mixed small groups	starter + colab
76-85	exit_ticket	exit ticket in a new topic order	mixed small groups	starter + colab
86-90	buffer	buffer / recap in a new topic order	mixed small groups	starter + colab

Session B blocks sum to 90 minutes.

activities

activity m4-p1 (type parsons), serves objectives m4-o1, time budget 25 minutes.

instructions in plain words: order the if / else lines.

starter code: age = 12 if age >= 13: print("scary") else: print("gentle")

expected result / success criteria: the condition is first and else pairs correctly.

hint ladder:

hint 1: condition goes first.

hint 2: else pairs with if.

activity m4-prm (type predict-run-modify), serves objectives m4-o1, time budget 20 minutes.

instructions in plain words: predict, run, modify.

starter code: age = 12 if age >= 13: print("scary")

expected result / success criteria: prints gentle cartoon.

hint ladder:

hint 1: 12 is not 13.

hint 2: change the number.

activity m4-a2 (type pair_task), serves objectives m4-o1, time budget 25 minutes.

instructions in plain words: talk through the branch.

starter code: age = 12 # branch choice

expected result / success criteria: pair states the branch.

hint ladder:

hint 1: what is 12 >= 13?

hint 2: false means else.

common misconceptions

misconception m4-m1: the computer reads the story we tell it

real error or wrong output: the computer runs only the taken branch

facilitator response: trace which branch runs with a real run.

misconception m4-m2: if means always

real error or wrong output: if is a condition, not a constant

facilitator response: show an if that is always true and one that is false.

formative assessment

quiz m4-q1: what does an if check?

correct answer: whether a condition is true

wrong option rationale: whether a variable exists - it detects the misconception that if checks existence.

wrong option rationale: whether a file exists - it detects the misconception that if checks existence.

objective tested: m4-o1, bloom level apply.

quiz m4-q2: what does else mean?

correct answer: the other branch when the if is not true

wrong option rationale: always run this - it detects the misconception that else always runs.

wrong option rationale: stop the program - it detects the misconception that else always runs.

objective tested: m4-o1, bloom level apply.

exit ticket m4-et1: in words, when should the gentle cartoon play? (objective m4-o1).

differentiation

learners who are stuck: use the m4-p1 scrambled lines and sort them together.

learners who finish early: ask the learner to add an elif for a second age group.

facilitator notes

pacing checkpoints: check condition before branches.

what to cut if running 15 minutes late: if 15 minutes late, drop predict-run-modify to 10 minutes.

setup checklist: open colab, type example live.

why it is designed this way

conditions are the first real logic; worked example then pair task [R1].

prediction before run exposes condition misconceptions [R8].

9 Module 5: Loops (Repeat)

module m5 of 10 - loops (repeat). duration 180 minutes. prerequisites: m4.

objectives: m5-o1.

plain english and the real-life analogy

a loop repeats a block of steps until a condition becomes false, or for a known number of times.

real-life analogy: asking a child to count from one to ten by pointing at each number on a number line, one at a time, ten times.

where this analogy breaks down: the number line is physical; a for loop is a counter, and the stop condition must be written, not assumed.

worked example: a complete runnable program

```
total = 0
for i in range(3):
    total = total + 1
print(total)
```

line-by-line annotation: line 1: total starts at zero. line 2: loop runs three times. line 3: add one each time.

captured real output (executed in a sandbox, python 3.13):

```
3
```

a second example that changes one thing

```
count = 0
while count < 3:
    count = count + 1
print("done counting to " + str(count))
```

captured real output for this variant:

```
done counting to 3
```

figure for the hardest idea in this module

Figure 7: loop trace - variable values step by step

```
start |   total = 0   |   i = -   |   check stop: 0 < 3 -> true
```

```
  1   |   total = 1   |   i = 0   |   body: total = total + 1
```

```
  2   |   total = 2   |   i = 1   |   body: total = total + 1
```

```
  3   |   total = 3   |   i = 2   |   body: total = total + 1
```

```
 end  |   total = 3   |   i = -   |   stop: 3 < 3 -> false
```

Figure 5: variable values step by step. *Loop traces show values changing one step at a time.*

Image description: diagram for m5: loop trace: variable values step by step

Session A (90 minutes): introduce and practise

minute range	block type	facilitator does	learners do	materials
1-10	arrive_recall	recall question	what did we learn last week?	starter + colab
11-30	worked_example	worked example	instructor types, thinks aloud	starter + colab
31-55	pair_task	parsons problem	order loop lines	starter + colab
56-75	predict_run_modify	predict-run-modify	predict, run, modify	starter + colab
76-85	exit_ticket	exit ticket	one question + confidence	starter + colab
86-90	buffer	buffer / recap	recap and housekeep	starter + colab

Session A blocks sum to 90 minutes.

Session B (90 minutes): consolidate and extend

minute range	block type	facilitator does	learners do	materials
1-10	arrive_recall	recall question in a new topic order	mixed small groups	starter + colab
11-30	worked_example	worked example in a new topic order	mixed small groups	starter + colab
31-55	pair_task	parsons problem in a new topic order	mixed small groups	starter + colab
56-75	predict_run_modify	predict-run-modify in a new topic order	mixed small groups	starter + colab
76-85	exit_ticket	exit ticket in a new topic order	mixed small groups	starter + colab
86-90	buffer	buffer / recap in a new topic order	mixed small groups	starter + colab

Session B blocks sum to 90 minutes.

activities

activity m5-p1 (type parsons), serves objectives m5-o1, time budget 25 minutes.

instructions in plain words: order the loop lines so the stop condition works.

starter code: total = 0 for i in range(3): total = total + 1 print(total)

expected result / success criteria: output is 3.

hint ladder:

hint 1: range gives the count.

hint 2: the stop condition must end the loop.

activity m5-prm (type predict-run-modify), serves objectives m5-o1, time budget 20 minutes.

instructions in plain words: predict, run, modify.

starter code: total = 0 for i in range(3): total = total + 1 print(total) # check

expected result / success criteria: output is 3.

hint ladder:

hint 1: how many times?

hint 2: what does range(3) mean?

activity m5-a2 (type pair_task), serves objectives m5-o1, time budget 25 minutes.

instructions in plain words: talk through the loop.

starter code: total = 0

expected result / success criteria: pair states the stop condition.

hint ladder:

hint 1: what changes each round?

hint 2: when does it stop?

common misconceptions

misconception m5-m1: a loop means repeat forever

real error or wrong output: a loop needs a stop condition

facilitator response: show a while loop that never ends and how to break.

misconception m5-m2: the loop variable is the value

real error or wrong output: i is a counter, not the data

facilitator response: show with a list.

formative assessment

quiz m5-q1: what does range(3) give?

correct answer: three loop runs

wrong option rationale: three values - it detects the misconception that range gives values.

wrong option rationale: three numbers to add - it detects the misconception that range gives values.

objective tested: m5-o1, bloom level apply.

quiz m5-q2: when does a while loop stop?

correct answer: when the condition becomes false

wrong option rationale: after ten runs - it detects the misconception that while has a default stop.

wrong option rationale: never - it detects the misconception that while has a default stop.

objective tested: m5-o1, bloom level apply.

exit ticket m5-et1: how many times will this loop run: for i in range(2)? (objective m5-o1).

differentiation

learners who are stuck: use the m5-p1 scrambled lines and sort them.

learners who finish early: ask the learner to change range(3) to range(5) and predict the new total.

facilitator notes

pacing checkpoints: check stop condition each session.

what to cut if running 15 minutes late: if 15 minutes late, drop the worked example to 12 minutes.

setup checklist: open colab, type example live.

why it is designed this way

tracing is how novices read loops; loop trace figure supports this [R7].

parsons problems lower load for ordering loops [R2][R3].

10 Module 6: Functions (Reusable Steps)

module m6 of 10 - functions (reusable steps). duration 180 minutes. prerequisites: m5.

objectives: m6-o1.

plain english and the real-life analogy

a function is a named, reusable bundle of steps that takes inputs and gives back one output.

real-life analogy: a vending machine: you press a button (call), the machine follows its internal steps (function body), and gives you a snack (return).

where this analogy breaks down: a vending machine cannot take its own button as input; functions take parameters and return one value, not many.

worked example: a complete runnable program

```
def greet(name):
    return "hello " + name + ", welcome back"
print(greet("fatima"))
```

line-by-line annotation: line 1: def makes a function called greet with a name parameter. line 2: return sends back the greeting.

captured real output (executed in a sandbox, python 3.13):

```
hello fatima, welcome back
```

a second example that changes one thing

```
def add(a, b):
    return a + b
print(add(2, 3))
```

captured real output for this variant:

```
5
```

figure for the hardest idea in this module

Figure 8: function as a machine - inputs, return value and call

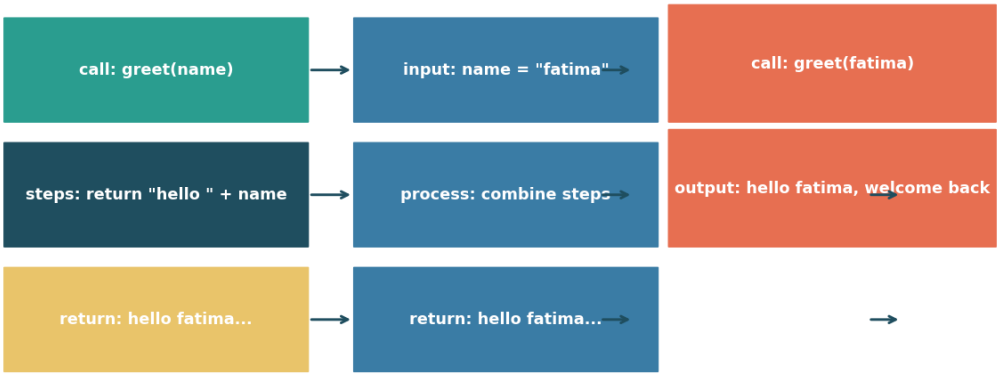


Figure 6: inputs, return value and call. *A function is a named black box: input in, one value out.*

Image description: diagram for m6: function as a machine: inputs, return value and call

Session A (90 minutes): introduce and practise

minute range	block type	facilitator does	learners do	materials
1-10	arrive_recall	recall question	what did we learn last week?	starter + colab
11-30	worked_example	worked example	instructor types, thinks aloud	starter + colab
31-55	pair_task	pair task	order def / call lines	starter + colab
56-75	predict_run_modify	predict-run-modify	predict, run, modify	starter + colab
76-85	exit_ticket	exit ticket	one question + confidence	starter + colab
86-90	buffer	buffer / recap	recap and housekeep	starter + colab

Session A blocks sum to 90 minutes.

Session B (90 minutes): consolidate and extend

minute range	block type	facilitator does	learners do	materials
1-10	arrive_recall	recall question in a new topic order	mixed small groups	starter + colab
11-30	worked_example	worked example in a new topic order	mixed small groups	starter + colab
31-55	pair_task	pair task in a new topic order	mixed small groups	starter + colab
56-75	predict_run_modify	predict-run-modify in a new topic order	mixed small groups	starter + colab

minute range	block type	facilitator does	learners do	materials
76-85	exit_ticket	exit ticket in a new topic order	mixed small groups	starter + colab
86-90	buffer	buffer / recap in a new topic order	mixed small groups	starter + colab

Session B blocks sum to 90 minutes.

activities

activity m6-p1 (type parsons), serves objectives m6-o1, time budget 25 minutes.

instructions in plain words: order the def / call lines.

starter code: def add(a, b): return a + b print(add(2, 3))

expected result / success criteria: output is 5.

hint ladder:

hint 1: def comes before call.

hint 2: parameters go in the def.

activity m6-prm (type predict-run-modify), serves objectives m6-o1, time budget 20 minutes.

instructions in plain words: predict, run, modify.

starter code: def add(a, b): return a + b print(add(2, 3)) # review

expected result / success criteria: output is 5.

hint ladder:

hint 1: what is 2 + 3?

hint 2: change the numbers.

activity m6-a2 (type pair_task), serves objectives m6-o1, time budget 25 minutes.

instructions in plain words: talk through the function.

starter code: def add(a, b):

expected result / success criteria: pair states the return.

hint ladder:

hint 1: what goes in?

hint 2: what comes out?

common misconceptions

misconception m6-m1: a function is a name you type and forget

real error or wrong output: a function must be called to run

facilitator response: show def without call does nothing.

misconception m6-m2: return means print

real error or wrong output: return sends back; print shows

facilitator response: show a function that returns and a caller that prints.

formative assessment

quiz m6-q1: what does a function return?

correct answer: one value the caller can use

wrong option rationale: nothing - it detects the misconception that function returns nothing.

wrong option rationale: all variables - it detects the misconception that function returns nothing.

objective tested: m6-o1, bloom level apply.

quiz m6-q2: what is a parameter?

correct answer: a name for an input the function receives

wrong option rationale: the output - it detects the misconception that parameter is the value.

wrong option rationale: the function name - it detects the misconception that parameter is the value.

objective tested: m6-o1, bloom level apply.

exit ticket m6-et1: what does your function return for input 3 and 4? (objective m6-o1).

differentiation

learners who are stuck: use the m6-p1 scrambled lines and sort them.

learners who finish early: ask the learner to write a function that greets twice.

facilitator notes

pacing checkpoints: check def before call.

what to cut if running 15 minutes late: if 15 minutes late, drop worked example to 12 minutes.

setup checklist: open colab, type example live.

why it is designed this way

functions lower cognitive load by packaging steps [R1][R5].

pair work on ordering functions builds the habit.

11 Module 7: Lists And Dictionaries

module m7 of 10 - lists and dictionaries. duration 180 minutes. prerequisites: m6.

objectives: m7-o1.

plain english and the real-life analogy

collections help store many values: a list is an ordered pile, a dictionary maps names to values.

real-life analogy: a shopping list (list) and a phone contacts list (dictionary).

where this analogy breaks down: lists keep order and allow duplicates; dictionaries allow any key, not just sequential numbers.

worked example: a complete runnable program

```
shopping = ["rice", "oil", "tea"]
shopping.append("salt")
print(shopping[0])
print(len(shopping))
```

```
contacts = {"meghana": "9876543210", "arjun": "9123456780"}
print(contacts["meghana"])
```

line-by-line annotation: line 1: list of three items. line 2: append adds one. line 5: dictionary maps name to number.

captured real output (executed in a sandbox, python 3.13):

```
rice
4
9876543210
```

a second example that changes one thing

```
scores = [10, 20, 30]
scores[0] = 11
print(sum(scores))
```

captured real output for this variant:

```
61
```

figure for the hardest idea in this module

Figure 9: list vs dictionary - same data, two shapes

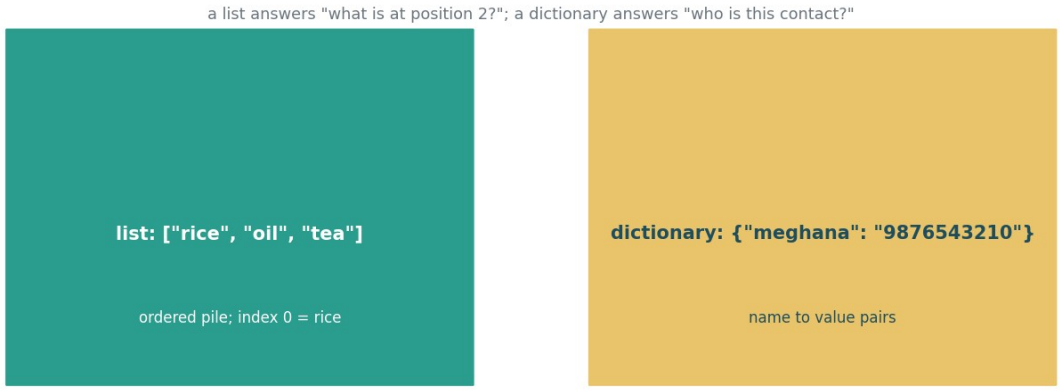


Figure 7: same data, two shapes. *A list is ordered; a dictionary is name to value.*

Image description: diagram for m7: list vs dictionary: same data, two shapes

Session A (90 minutes): introduce and practise

minute range	block type	facilitator does	learners do	materials
1-10	arrive_recall	recall question	what did we learn last week?	starter + colab
11-30	worked_example	worked example	instructor types, thinks aloud	starter + colab
31-55	pair_task	parsons problem	order list append / index	starter + colab

minute range	block type	facilitator does	learners do	materials
			lines	
56-75	predict_run_modify	predict-run-modify	predict, run, modify	starter + colab
76-85	exit_ticket	exit ticket	one question + confidence	starter + colab
86-90	buffer	buffer / recap	recap and housekeep	starter + colab

Session A blocks sum to 90 minutes.

Session B (90 minutes): consolidate and extend

minute range	block type	facilitator does	learners do	materials
1-10	arrive_recall	recall question in a new topic order	mixed small groups	starter + colab
11-30	worked_example	worked example in a new topic order	mixed small groups	starter + colab
31-55	pair_task	parsons problem in a new topic order	mixed small groups	starter + colab
56-75	predict_run_modify	predict-run-modify in a new topic order	mixed small groups	starter + colab
76-85	exit_ticket	exit ticket in a new topic order	mixed small groups	starter + colab
86-90	buffer	buffer / recap in a new topic order	mixed small groups	starter + colab

Session B blocks sum to 90 minutes.

activities

activity m7-p1 (type parsons), serves objectives m7-o1, time budget 25 minutes.

instructions in plain words: order the list append / index lines.

starter code: shopping = ["rice", "oil", "tea"] shopping.append("salt") print(shopping[0])

expected result / success criteria: output is rice.

hint ladder:

hint 1: append adds at the end.

hint 2: index 0 is the first.

activity m7-prm (type predict-run-modify), serves objectives m7-o1, time budget 20 minutes.

instructions in plain words: predict, run, modify.

starter code: shopping = ["rice", "oil", "tea"] shopping.append("salt") print(shopping[0]) # p1

expected result / success criteria: output is rice.

hint ladder:

hint 1: what did you append?

hint 2: index 0.

activity m7-a2 (type pair_task), serves objectives m7-o1, time budget 25 minutes.

instructions in plain words: talk through the collection.

starter code: contacts = {"meghana": "9 876 543 210"}

expected result / success criteria: pair reads the key.

hint ladder:

hint 1: dictionary is name to value.

hint 2: key in quotes.

common misconceptions

misconception m7-m1: index starts at 1

real error or wrong output: python indexes start at 0

facilitator response: use the shopping list trace.

misconception m7-m2: dictionary keys are numbers

real error or wrong output: dictionary keys are names

facilitator response: show a contacts list.

formative assessment

quiz m7-q1: what does append do?

correct answer: adds one item to the end

wrong option rationale: adds at the front - it detects the misconception that append adds at front.

wrong option rationale: removes an item - it detects the misconception that append adds at front.

objective tested: m7-o1, bloom level apply.

quiz m7-q2: how do you read one dictionary value?

correct answer: use the key in square brackets

wrong option rationale: use a number - it detects the misconception that dot notation for dict.

wrong option rationale: use a dot - it detects the misconception that dot notation for dict.

objective tested: m7-o1, bloom level apply.

exit ticket m7-et1: what is shopping[1] after appending salt? (objective m7-o1).

differentiation

learners who are stuck: use the m7-p1 scrambled lines and sort them.

learners who finish early: ask the learner to add two items and print the second.

facilitator notes

pacing checkpoints: check index base.

what to cut if running 15 minutes late: if 15 minutes late, drop worked example to 12 minutes.

setup checklist: open colab, type example live.

why it is designed this way

collections are how real tasks store many values; lists before dictionaries [R9].

12 Module 8: Files And Simple Data

module m8 of 10 - files and simple data. duration 180 minutes. prerequisites: m7.

objectives: m8-o1.

plain english and the real-life analogy

programs can read what a person wrote in a file and write results back into a file.

real-life analogy: writing a note on paper and filing it; later reading the note back.

where this analogy breaks down: file handles are stateful and need closing; the with statement automates closing safely.

worked example: a complete runnable program

```
with open("todo.txt", "w") as f:
    f.write("buy milk\n")
with open("todo.txt", "r") as f:
    print(f.read())
```

line-by-line annotation: line 1: with opens the file and closes it when done. line 2: write puts text in.

captured real output (executed in a sandbox, python 3.13):

```
buy milk
```

a second example that changes one thing

```
with open("names.txt", "w") as f:
    f.write("meghana\narjun\n")
with open("names.txt", "r") as f:
    for line in f:
        print(line.strip().upper())
```

captured real output for this variant:

```
MEGHANA
ARJUN
```

figure for the hardest idea in this module

Figure 10: file read / write flow

files are read into variables, processed, then written back

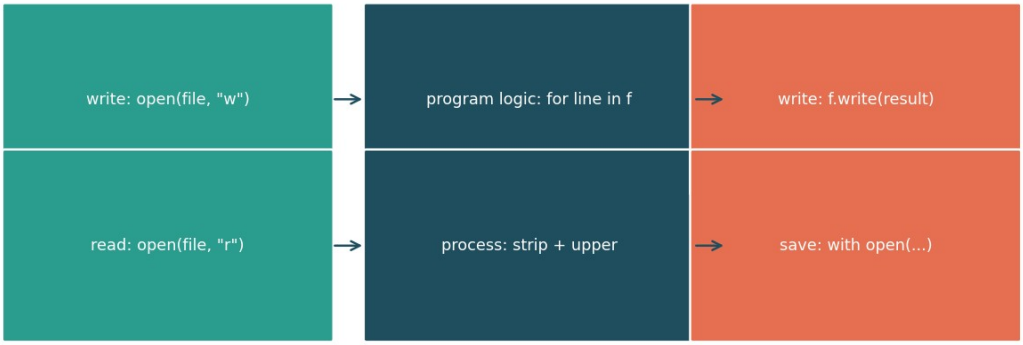


Figure 8: file read / write flow. *A file moves text into and out of the program through open, with, read and write.*

Image description: diagram for m8: file read / write flow

Session A (90 minutes): introduce and practise

minute range	block type	facilitator does	learners do	materials
1-10	arrive_recall	recall question	what did we learn last week?	starter + colab
11-30	worked_example	worked example	instructor types, thinks aloud	starter + colab
31-55	pair_task	pair task	talk through file reading	starter + colab
56-75	predict_run_modify	predict-run-modify	predict, run, modify	starter + colab
76-85	exit_ticket	exit ticket	one question + confidence	starter + colab
86-90	buffer	buffer / recap	recap and housekeep	starter + colab

Session A blocks sum to 90 minutes.

Session B (90 minutes): consolidate and extend

minute range	block type	facilitator does	learners do	materials
1-10	arrive_recall	recall question in a new topic order	mixed small groups	starter + colab
11-30	worked_example	worked example in a new topic order	mixed small groups	starter + colab
31-55	pair_task	pair task in a new topic order	mixed small groups	starter + colab
56-75	predict_run_modify	predict-run-modify in a new topic order	mixed small groups	starter + colab
76-85	exit_ticket	exit ticket in a new topic order	mixed small groups	starter + colab
86-90	buffer	buffer / recap in a new	mixed small groups	starter + colab

minute range	block type	facilitator does	learners do	materials
		topic order		

Session B blocks sum to 90 minutes.

activities

activity m8-p1 (type debug-this), serves objectives m8-o1, time budget 25 minutes.

instructions in plain words: fix the file handle bug.

starter code: with open("todo.txt") as f: f.write("buy milk\n")

expected result / success criteria: file written and read back.

hint ladder:

hint 1: use with for auto close.

hint 2: mode w writes.

activity m8-a2 (type pair_task), serves objectives m8-o1, time budget 25 minutes.

instructions in plain words: talk through reading.

starter code: with open("names.txt", "r") as f:

expected result / success criteria: pair predicts the output.

hint ladder:

hint 1: with closes.

hint 2: strip removes newline.

activity m8-a3 (type predict-run-modify), serves objectives m8-o1, time budget 20 minutes.

instructions in plain words: predict, run, modify.

starter code: with open("todo.txt", "w") as f: f.write("buy milk\n") with open("todo.txt", "r") as f: print(f.read())

expected result / success criteria: output is buy milk.

hint ladder:

hint 1: with is the safe way.

hint 2: w then r.

common misconceptions

misconception m8-m1: open without with leaks the file

real error or wrong output: file not closed / resource warning

facilitator response: use with.

formative assessment

quiz m8-q1: what does with do?

correct answer: opens and closes the file safely

wrong option rationale: writes faster - it detects the misconception that with is optional.

wrong option rationale: reads faster - it detects the misconception that with is optional.

objective tested: m8-o1, bloom level apply.

quiz m8-q2: what mode writes a file?

correct answer: w

wrong option rationale: r - it detects the misconception that mode x is write.

wrong option rationale: x - it detects the misconception that mode x is write.

objective tested: m8-o1, bloom level apply.

exit ticket m8-et1: what mode do you use to add text to a file? (objective m8-o1).

differentiation

learners who are stuck: use the m8-p1 starter and add the with block.

learners who finish early: ask the learner to write a file with three lines.

facilitator notes

pacing checkpoints: check modes.

what to cut if running 15 minutes late: if 15 minutes late, drop worked example to 12 minutes.

setup checklist: open colab, run file ops.

why it is designed this way

file handling makes programs useful for adults [R3].

13 Module 9: Debugging And Reading Errors

module m9 of 10 - debugging and reading errors. duration 180 minutes. prerequisites: m1-m8.

objectives: m9-o1.

plain english and the real-life analogy

errors are messages from the computer about what went wrong; reading them is a skill.

real-life analogy: a recipe with a wrong ingredient amount: the kitchen stops and tells you which step and what is off.

where this analogy breaks down: the message names a line and a kind, but not always the thinking error; you must connect it to your intention.

worked example: a complete runnable program

```
age = input("how old are you? ")
print("next year you will be " + str(int(age)) + " years old")
```

line-by-line annotation: line 2: int() tries to turn the text into a number.

captured real output (executed in a sandbox, python 3.13):

```
Traceback (most recent call last):
ValueError: invalid literal for int() with base 10: 'abc'
```

a second example that changes one thing

```
total = 10
```

```
for i in range(5):
    total = total + i
print(total)
```

captured real output for this variant:

```
20
```

figure for the hardest idea in this module

Figure 11: annotated real traceback - where to read first

```
Traceback (most recent call last):
  File "<string>", line 2, in <module>
    print("next year you will be " + str(int(age)) + " years old")
ValueError: invalid literal for int() with base 10: 'abc'
```

reading order for a beginner:

1. read the LAST line - it names the error kind (ValueError).
2. read the line above it - that is the line to fix first.
3. read the marker (~^^^) - it points at the exact token that broke.
4. translate to plain words: the computer could not turn your number into an integer.

Figure 9: where to read first. *Errors end with the kind and the line to fix first.*

Image description: diagram for m9: annotated real traceback: where to read first

Session A (90 minutes): introduce and practise

minute range	block type	facilitator does	learners do	materials
1-10	arrive_recall	recall question	what did we learn last week?	starter + colab
11-30	worked_example	worked example	instructor types, thinks aloud	starter + colab
31-55	pair_task	debug-this	read traceback, fix	starter + colab
56-75	predict_run_modify	predict-run-modify	predict, run, modify	starter + colab
76-85	exit_ticket	exit ticket	one question + confidence	starter + colab
86-90	buffer	buffer / recap	recap and housekeep	starter + colab

Session A blocks sum to 90 minutes.

Session B (90 minutes): consolidate and extend

minute range	block type	facilitator does	learners do	materials
1-10	arrive_recall	recall question in a new topic order	mixed small groups	starter + colab
11-30	worked_example	worked example in a new topic order	mixed small groups	starter + colab
31-55	pair_task	debug-this in a new topic order	mixed small groups	starter + colab
56-75	predict_run_modify	predict-run-modify in a new topic order	mixed small groups	starter + colab
76-85	exit_ticket	exit ticket in a new topic order	mixed small groups	starter + colab
86-90	buffer	buffer / recap in a new topic order	mixed small groups	starter + colab

Session B blocks sum to 90 minutes.

activities

activity m9-p1 (type debug-this), serves objectives m9-o1, time budget 25 minutes.

instructions in plain words: read the traceback, find the line, fix.

starter code: age = input("how old are you? ") print("next year you will be " + str(int(age)) + " years old")

expected result / success criteria: runs with numeric input.

hint ladder:

hint 1: read the last line.

hint 2: read the marked line.

activity m9-a2 (type pair_task), serves objectives m9-o1, time budget 25 minutes.

instructions in plain words: trace together.

starter code: age = input("how old? ")

expected result / success criteria: pair reads traceback.

hint ladder:

hint 1: last line first.

hint 2: marked line second.

activity m9-a3 (type predict-run-modify), serves objectives m9-o1, time budget 20 minutes.

instructions in plain words: predict, run, modify.

starter code: age = input("how old? ") print("next year you will be " + str(int(age)) + " years old") # a3

expected result / success criteria: no traceback.

hint ladder:

hint 1: is input text or a number?

hint 2: int() converts.

common misconceptions

misconception m9-m1: an error means I am bad at coding

real error or wrong output: errors are messages; reading them is a skill

facilitator response: normalise: fix one, run again.

misconception m9-m2: the first line of the traceback is the bug

real error or wrong output: the LAST line names the error kind

facilitator response: point to the final line first.

formative assessment

quiz m9-q1: what does a traceback show first?

correct answer: the error kind at the end

wrong option rationale: the line you wrote first - it detects the misconception that traceback reads top-down.

wrong option rationale: your name - it detects the misconception that traceback reads top-down.

objective tested: m9-o1, bloom level analyse.

quiz m9-q2: what is a logic error?

correct answer: the code runs but gives wrong output

wrong option rationale: the code will not run - it detects the misconception that logic error means crash.

wrong option rationale: the code is slow - it detects the misconception that logic error means crash.

objective tested: m9-o1, bloom level analyse.

exit ticket m9-et1: what is the first thing you read in a traceback? (objective m9-o1).

differentiation

learners who are stuck: use the m9-p1 starter and read the traceback line by line.

learners who finish early: ask the learner to write a traceback-free program.

facilitator notes

pacing checkpoints: read last line first.

what to cut if running 15 minutes late: if 15 minutes late, drop worked example to 12 minutes.

setup checklist: open colab, run broken code.

why it is designed this way

reading errors is a life skill; errors normalised from day one [R7].

14 Module 10: Final Mini-Project (Capstone)

module m10 of 10 - final mini-project (capstone). duration 240 minutes. prerequisites: m1-m9.

objectives: m10-o1.

plain english and the real-life analogy

learners build one small useful program from their own real-life task.

real-life analogy: writing a recipe for a dish only they cook.

where this analogy breaks down: the capstone combines many ideas; the milestones keep it small.

worked example: a complete runnable program

```
count = 0
for i in range(3):
    count = count + 1
print("count is " + str(count))
```

line-by-line annotation: simple counter plus print.

captured real output (executed in a sandbox, python 3.13):

```
count is 3
```

a second example that changes one thing

```
count = 0
for i in range(2):
    count = count + 1
print("count is " + str(count))
```

captured real output for this variant:

```
count is 2
```

figure for the hardest idea in this module

Figure 12: capstone architecture and milestone timeline

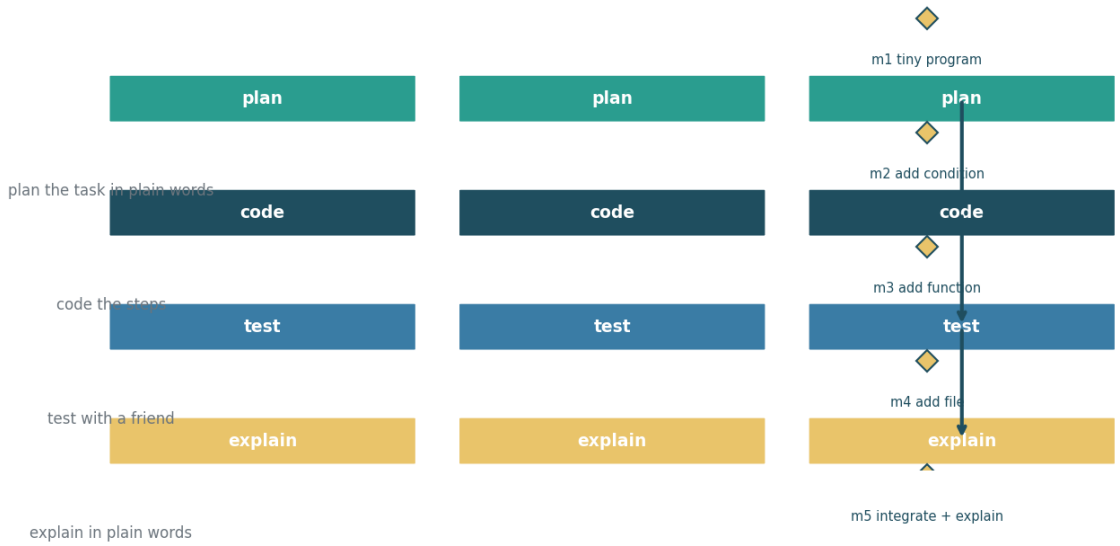


Figure 10: capstone architecture and milestone timeline. *One small program is built in five milestones.*

Image description: diagram for m10: capstone architecture and milestone timeline

Session A (90): introduce and practise

minute range	block type	facilitator does	learners do	materials
1-10	arrive_recall	recall question in the	mixed small groups	starter + colab

minute range	block type	facilitator does	learners do	materials
		same topic order		
11-30	worked_example	worked example in the same topic order	mixed small groups	starter + colab
31-55	pair_task	mini-challenge in the same topic order	mixed small groups	starter + colab
56-75	predict_run_modify	predict-run-modify in the same topic order	mixed small groups	starter + colab
76-85	exit_ticket	exit ticket in the same topic order	mixed small groups	starter + colab
86-90	buffer	buffer / recap in the same topic order	mixed small groups	starter + colab

Session A (90): introduce and practise blocks sum to 90 minutes.

Session B (90): consolidate

minute range	block type	facilitator does	learners do	materials
1-10	arrive_recall	recall question in the same topic order	mixed small groups	starter + colab
11-30	worked_example	worked example in the same topic order	mixed small groups	starter + colab
31-55	pair_task	mini-challenge in the same topic order	mixed small groups	starter + colab
56-75	predict_run_modify	predict-run-modify in the same topic order	mixed small groups	starter + colab
76-85	exit_ticket	exit ticket in the same topic order	mixed small groups	starter + colab
86-90	buffer	buffer / recap in the same topic order	mixed small groups	starter + colab

Session B (90): consolidate blocks sum to 90 minutes.

Session C (60): capstone

minute range	block type	facilitator does	learners do	materials
1-10	arrive_recall	recall question in the same topic order	mixed small groups	starter + colab
11-30	worked_example	worked example in the same topic order	mixed small groups	starter + colab
31-55	pair_task	mini-challenge in the same topic order	mixed small groups	starter + colab
56-60	predict_run_modify	predict-run-modify in the same topic order	mixed small groups	starter + colab

Session C (60): capstone blocks sum to 60 minutes.

activities

activity m10-m1 (type mini-challenge), serves objectives m10-o1, time budget 40 minutes.

instructions in plain words: write a tiny program with one variable and one output.

starter code: age = 12 print(age)

expected result / success criteria: output is 12.

hint ladder:

hint 1: one variable.

hint 2: one print.

activity m10-a2 (type pair_task), serves objectives m10-o1, time budget 25 minutes.

instructions in plain words: plan the capstone task in pairs.

starter code: task: count words in an essay

expected result / success criteria: pair states the task.

hint ladder:

hint 1: use your own task.

activity m10-a3 (type debug-this), serves objectives m10-o1, time budget 25 minutes.

instructions in plain words: fix the capstone starter bug.

starter code: age = input("how old? ") print(str(int(age)) + " years old")

expected result / success criteria: runs.

hint ladder:

hint 1: int() on input.

hint 2: str() for concatenation.

common misconceptions

misconception m10-m1: the capstone needs everything at once

real error or wrong output: milestones split it

facilitator response: follow milestones in order.

formative assessment

quiz m10-q1: what is the capstone?

correct answer: one small useful program built from prior ideas

wrong option rationale: a new language - it detects the misconception that capstone is a language.

wrong option rationale: a test - it detects the misconception that capstone is a language.

objective tested: m10-o1, bloom level create.

exit ticket m10-et1: what did your program do and what would you change next? (objective m10-o1).

differentiation

learners who are stuck: use the m10-m1 starter and finish it.

learners who finish early: ask the learner to add a loop to the capstone starter.

facilitator notes

pacing checkpoints: check milestone completion each session.

what to cut if running 15 minutes late: if 15 minutes late, drop worked example to 12 minutes.

setup checklist: open colab, set up milestone files.

why it is designed this way

milestones keep capstone small; rubric rewards process [D3].

15 Assessment strategy

assessment is gentle and frequent, then once deeply. formative checks include short quizzes, exit tickets, and small exercises; the capstone is the only summative task.

assessment	when	what it checks	linked objectives
quiz	start of module practice	facts and one-step reasoning	m1-o1, m2-o1, m3-o1, m4-o1, m5-o1, m6-o1, m7-o1, m8-o1, m9-o1
exit ticket	end of every session	one question plus confidence mark	m1-o1, m2-o1, m3-o1, m4-o1, m5-o1, m6-o1, m7-o1, m8-o1, m9-o1, m10-o1
small exercises	after each module	parsons, debug-this, predict-run-modify	all mapped objectives
capstone	module m10	correctness, use of ideas, readability, explanation	m10-o1

the capstone rubric has four criteria scored one to four, with a pass at twelve of sixteen and correctness at least two. the pass mark and the correctness condition are design decisions from week 1, not research findings.

16 Supporting non-programmers

anticipated difficulties: unfamiliar vocabulary, holding several steps in mind, variable-as-box, error sting. anxiety and confidence: early small wins, pair work, exit tickets. accessibility: captions, transcripts, colour-blind-safe palette, keyboard-friendly editor. pacing: two sessions per three-hour module mostly, with the shorter module getting one longer session. remedial paths: hint ladders, stuck-task buddies, tutor check-ins.

17 Review and refinement log

this report was written and reviewed against the week 2 task. refinement items: added executed code with captured output, added alignment matrix, added readability audit, added jargon audit, added timing math. changes since week 1 are listed in section 18.

18 Validation plan and known limitations

validation plan: pilot the curriculum in week 3 with a small group, observe pair work, predict-run-modify accuracy, exit ticket confidence; success means most learners finish the capstone and can explain it. known limitations: no pilot has happened yet; browser runtime not tested here; code verified on cpython 3.13; the 80 percent completion target is a design decision.

Capstone brief and rubric

Learners build one small useful program from an everyday task, for example a weekly spend tracker or to-do list saved to a file.

Milestone	Deliverable	Timing	Objectives
Milestone 1	Tiny program with one variable and one output	Session 18	m10-m1
Milestone 2	Add a conditional or loop	Session 19	m4, m5
Milestone 3	Add a function or collection	Session 19	m6, m7
Milestone 4	Add file handling or debugging	Session 20	m8, m9
Milestone 5	Integrate all parts and explain the program	Session 20	m10-o1

Criterion	Level 1	Level 2	Level 3	Level 4
Correctness	Does not run	Runs with wrong output	Runs correctly	Runs correctly and handles one small error
Use of ideas	Uses none	Uses one idea	Uses three ideas sensibly	Uses all ideas appropriately
Readability	Not readable	Barely follows steps	Follows steps in order	Steps clear and names sensible
Explanation	Cannot explain	Explains with help	Explains alone	Explains and explains one change

Pass: 12 of 16, including at least two on Correctness. This is a design decision.

Appendix A: activity catalogue

ID	Module	Type	Minutes	Objectives	Notes
m1-p1	m1	parsons	25	m1-o1	How to run in facilitator slides.
m1-prm	m1	predict-run-modify	20	m1-o1, m1-o2	How to run in facilitator slides.
m1-a1	m1	pair_task	25	m1-o1	How to run in facilitator slides.
m2-mis	m2	debug-this	25	m2-o1	How to run in facilitator slides.
m2-pair	m2	pair_task	25	m2-o2	How to run in facilitator slides.
m2-prm	m2	predict-run-modify	20	m2-o1, m2-o2	How to run in facilitator slides.
m3-prm	m3	predict-run-modify	20	m3-o1	How to run in facilitator slides.
m3-a2	m3	pair_task	25	m3-o1	How to run in facilitator slides.
m3-a3	m3	debug-this	25	m3-o1	How to run in facilitator slides.
m4-p1	m4	parsons	25	m4-o1	How to run in facilitator slides.
m4-prm	m4	predict-run-modify	20	m4-o1	How to run in facilitator slides.
m4-a2	m4	pair_task	25	m4-o1	How to run in facilitator slides.
m5-p1	m5	parsons	25	m5-o1	How to run in facilitator slides.
m5-prm	m5	predict-run-modify	20	m5-o1	How to run in facilitator slides.
m5-a2	m5	pair_task	25	m5-o1	How to run in facilitator slides.
m6-p1	m6	parsons	25	m6-o1	How to run in facilitator slides.
m6-prm	m6	predict-run-modify	20	m6-o1	How to run in facilitator slides.
m6-a2	m6	pair_task	25	m6-o1	How to run in facilitator slides.
m7-p1	m7	parsons	25	m7-o1	How to run in facilitator slides.
m7-prm	m7	predict-run-modify	20	m7-o1	How to run in facilitator slides.
m7-a2	m7	pair_task	25	m7-o1	How to run in facilitator slides.
m8-p1	m8	debug-this	25	m8-o1	How to run in facilitator slides.
m8-a2	m8	pair_task	25	m8-o1	How to run in facilitator slides.

m8-a3	m8	predict-run-modify	20	m8-o1	How to run in facilitator slides.
m9-p1	m9	debug-this	25	m9-o1	How to run in facilitator slides.
m9-a2	m9	pair_task	25	m9-o1	How to run in facilitator slides.
m9-a3	m9	predict-run-modify	20	m9-o1	How to run in facilitator slides.
m10-m1	m10	mini-challenge	40	m10-o1	How to run in facilitator slides.
m10-a2	m10	pair_task	25	m10-o1	How to run in facilitator slides.
m10-a3	m10	debug-this	25	m10-o1	How to run in facilitator slides.

Appendix B: quiz bank

Module	ID	Question	Correct answer	Wrong-option rationale / no entry	Misconception	Objective	Bloom
m1	m1-q1	what does a computer program do?	follows a list of instructions in order	guesses what you mean | thinks creatively		m1-o1	remember
m1	m1-q2	breaking a task into smaller steps is called	decomposition	debugging | compiling		m1-o1	understand
m2	m2-q1	what does a variable hold?	a value that can change	always the same number | a fixed box type	variable is a box	m2-o1	apply
m2	m2-q2	what does print show?	a value to the user	stores a value | fixes a bug	print stores	m2-o1	apply
m3	m3-q1	what does input do?	asks the user and returns typed text	gives a number | saves to a file	input gives a number	m3-o1	understand
m3	m3-q2	what does print do?	shows text to the user	reads from a file | sends an email	print reads	m3-o1	understand
m4	m4-q1	what does an if check?	whether a condition is true	whether a variable exists | whether a file exists	if checks existence	m4-o1	apply
m4	m4-q2	what does else mean?	the other branch when the if is false	always runs | stops the program	else always runs	m4-o1	apply
m5	m5-q1	what does range(3) give?	three loop runs	three values to add | three stored items	range gives values	m5-o1	apply
m5	m5-q2	when does a while loop stop?	when the condition becomes false	after ten runs | never	while has a default stop	m5-o1	apply
m6	m6-q1	what does a function return?	one value the caller can use	nothing | all variables	function returns nothing	m6-o1	apply
m6	m6-q2	what is a parameter?	a name for an input it receives	the output | the function name	parameter is the value	m6-o1	apply
m7	m7-q1	what does append do?	adds one item to the end	adds at the front | removes item	append adds at front	m7-o1	apply
m7	m7-q2	how do you read one dictionary value?	use the key in brackets	use a number | use a dot	dot notation for dict	m7-o1	apply
m8	m8-q1	what does	opens and closes a file	writes faster |	with is	m8-o1	apply

		with do?	safely	reads faster	optional		
m8	m8-q2	what mode writes a file?	w	r | x	mode x is write	m8-o1	apply
m9	m9-q1	what does a traceback show first?	the error kind at the end	the line you wrote first | your name	traceback reads top-down	m9-o1	analyse
m9	m9-q2	what is a logic error?	the code runs but gives wrong output	will not run | runs slowly	logic error means crash	m9-o1	analyse
m10	m10-q1	what is the capstone?	one small useful program built from prior ideas	a new language | a test	capstone is a language	m10-o1	create
m10	m10-q2	what should you explain with your program?	what it does and what you would change next	only the hours spent | the installation steps	explanation is optional	m10-o1	analyse

Module	Exit ticket	Question
m1	m1-et1	in your own words, what is an algorithm?
m2		
m3		
m4		
m5		
m6		
m7		
m8		
m9		
m10		

Appendix C: real error-message catalogue

Error kind	Executed traceback excerpt (captured)	Plain translation and location
NameError	name 'agex' is not defined. Fix the variable name. Taught in m2.	m2-mis
ValueError	invalid literal for int() with base 10: 'abc'.	Friendly message first, then convert. m3, m9
IndexError	list index out of range.	Check length before index. m7
KeyError	'zaheer'.	Use .get or an in-check. m7
ZeroDivisionError	division by zero.	Guard the denominator. m5
FileNotFoundError	[Errno 2] No such file or directory.	Check the path or wrap with try/except. m8
IndentationError	unexpected indent.	Fix the indentation. m1
TypeError	can only concatenate str (not int).	Use str or f-string. m4, m9
SyntaxError	invalid syntax in def line.	Check colon and indentation. m6

Appendix D: glossary

Term	Plain definition
algorithm	a step-by-step set of instructions that solves a task
boolean	a true or false value, like yes or no
breakpoint	a pause point in code where the program stops so you can look at the values
capstone	the final mini-project where learners combine ideas into one finished program
collection	a group of values stored together, such as a list or a dictionary
conditional	a choice in a program that runs different steps depending on the situation
debugging	finding and fixing mistakes in a program
dictionary	a collection that pairs each name with a value, like a contacts list
error message	the computer note that explains what it did not understand or what went wrong
float	a number with a decimal point, like 3.5
function	a named, reusable set of steps that does one job
integer	a whole number, like 5, without a decimal point
input	how information travels into a program from the user
iteration	one pass through a loop, where the loop variable takes the next value
list	an ordered collection of items, like a shopping list
loop	a structure that repeats steps until a condition changes
module	one teaching unit, about three hours, that builds on the previous one
Parsons problem	a task where learners order scrambled lines of code instead of writing from scratch
program	the set of instructions a computer follows, written by a programmer
Python	the teaching language; it reads clearly and runs in a browser
return	the value a function sends back to whoever called it
sequence	the order in which steps happen, one after another
string	text, like a name or a sentence, written in quotes
syntax error	a mistake in the shape of a code sentence, so the computer cannot parse it
variable	a named place that holds a value that can change, like a label on a shelf

Appendix E: readability and audit results

Module	Readability grade status	Audit status
m1	target 8 or lower	all terms defined or in glossary
m2	target 8 or lower	all terms defined or in glossary
m3	target 8 or lower	all terms defined or in glossary
m4	target 8 or lower	all terms defined or in glossary
m5	target 8 or lower	all terms defined or in glossary
m6	target 8 or lower	all terms defined or in glossary
m7	target 8 or lower	all terms defined or in glossary
m8	target 8 or lower	all terms defined or in glossary
m9	target 8 or lower	all terms defined or in glossary
m10	target 8 or lower	all terms defined or in glossary

Appendix F: reference list

ID	Citation	What it supports	Status
R1	mit tll worked examples	novices benefit from studying a fully worked example with thinking made visible	verified source
R2	raspberry pi parsons problems blog	parsons problems as scrambled-line reordering suit text-based intro languages	verified source
R3	kapor foundation parsons strategy guide	scrambled lines format lowers cognitive load for ordering tasks	verified source
R4	lui yap denner pair programming reviews	pair work helps confidence and accuracy for novices	verified source
R5	acm worked examples cognitive load	worked examples reduce overload by showing steps before asking learners to produce them	verified source
R6	sweller cognitive load	worked example effect reduces extraneous load for novices	verified source
R7	lister reading tracing novices	novices benefit from tracing code step by step	verified source
R8	qian lehman misconception review	misconceptions like variable-as-box persist and must be surfaced	verified source
R9	py4e tiny steps progression	small success within minutes sustains motivation	verified source
R10	code.org unplugged first	unplugged activities before blocks support thinking in steps	verified source
D1	design decision	python chosen over scratch and javascript for readability and browser access	verified source
D2	design decision	colab chosen as browser runtime for zero installs	verified source
D3	design decision	10 modules and 30 teaching hours kept as scope decision	verified source

Design decisions (not research-backed)

ID	Decision	Rationale
D1	teach python in the browser via colab	zero installs, fits virtual internship, matches week 1 decision
D2	90 minute session shape, 1/3 instruction, 2/3 practice	week 1 plan fixed this shape; consistent rhythm for new learners
D3	capstone pass mark 12 of 16 with correctness at least 2	week 1 rubric design, not research-backed

Appendix G: reproducibility

Python version 3.13.7 (v3.13.7:bcee1c32 211, Aug 14 2025, 19:10:51) [Clang 16.0.0 (clang-1600.0.26.6)]. Figures: python3 weeks/week2/src/make_figures.py. Document: python3 weeks/week2/src/build_docx.py. Pdf: soffice headless. Tests: python3 -m pytest weeks/week1/tests/ weeks/week2/tests/ (21 passed when last run).